1064 Center Dr
Gainesville, FL 32611
February 22, 2026

Dr. Kurt Schulze
University of Florida
1064 Center Dr
Gainesville, FL 32611

Dear Dr. Schulze:

The objective of this experiment is to evaluate the performance characteristics of two centrifugal fans with different impeller blade geometry—one with radial blades and the other with backwards-inclined blades—operating at multiple speeds. Specifically, the lab aims to determine the relationship between head rise, volumetric flow rate, shaft power and efficiency for both fans. Dimensionless performance coefficients corresponding to head, power and flow will be calculated to assess similarity behavior between the two fans operating at varying speeds and air flow conditions for each data set.

The theoretical foundation of this experiment is based on turbomachinery energy transfer and similarity principles. The head rise across the fan was determined from the measured pressure difference divided by the product of air density and the gravitational constant. Air density was found beforehand using the ideal gas law based on the measured atmospheric pressure, ambient temperature, and the specific gas constant for air. The volumetric flow rate was calculated from the measured nozzle pressure differential using the calibrated nozzle flow equation. This relation is based on conservation of mass and energy principles and incorporates a discharge coefficient to account for non-ideal effects such as viscous losses and flow contraction. Brake horsepower (BHP), or shaft work, was computed from measured torque and impeller speed, while the water horsepower (WHP), or work imparted to the fluid, was determined from the product of head rise, flow rate, air density and the gravitational constant. Nondimensional coefficients were formed using impeller diameter, rotational speed and air density to collapse performance data according to turbomachinery affinity laws which predict that flow rate scales linearly with rotational speed, head scales with speed squared, and power scales with speed cubed.

The experimental apparatus consisted of two MFP106 centrifugal fan rigs equipped with interchangeable impellers, a dynamometer paired with an electric motor to measure speed and torque, pressure taps upstream and downstream of the fan, and a calibrated flow measurement nozzle. Data collection was performed through a Versatile Data Acquisition System (VDAS) integrated with LabVIEW to record pressures, torque, rotational speed and total power. Beginning with the radial blades impeller fan, five rotational speeds increasing from 1000 rpm to 3000 rpm by 500 rpm increments were tested for eleven different flow rates. These flow rates varied from 100% to 0% and were decreased by 10% increments using a slider plate that adjusted a downstream valve. A specific mechanical power loss was entered into the software for each new fan speed and data acquisition was set to take 5 seconds with 0.5 second intervals, resulting in eleven data points for every set. Approximately five seconds were waited between each set before recording data to ensure the flow measurements steadied out after adjusting the slider plate. Once all data sets were collected for the radial blades impeller fan, the results were saved and the same procedure was then carried out for the backwards-inclined blades impeller fan.

The experimental results closely followed expected turbomachinery behavior. For each rotational speed, the head rise slowly increased with flow rate until approximately the seventh data point, after which it began to decrease due to greater aerodynamic and frictional losses at higher flow rates. For example, at 3000 rpm for the radial impeller, head rose to just over 100 meters before gradually declining at higher flow rates, while at 1000 rpm the peak head rise was only about 10 meters. Increasing rotational speed shifted the performance curves upward, with maximum flow increasing from approximately 0.08 m^3/s at 1000 rpm to about 0.24 m^3/s at 3000 rpm, consistent with affinity law scaling. Water horsepower and brake horsepower increased nearly linearly with flow rate for both fans, and at higher speeds the gap between WHP and BHP narrowed, indicating improved efficiency. The backwards-inclined impeller achieved the highest peak efficiency around 80% for the 3000 rpm data set, while the radial impeller maximum efficiency converged at 70%. Nondimensional performance coefficients for head, power and flow showed reasonable curve collapse, confirming similarity behavior, with slightly lower values for the backwards-inclined design due to blade geometry physics.

Sincerely,

Bryce Addeo | Sean Muller | Matthew Strogach

"On my honor, I have neither given nor received unauthorized aid in doing this assignment."

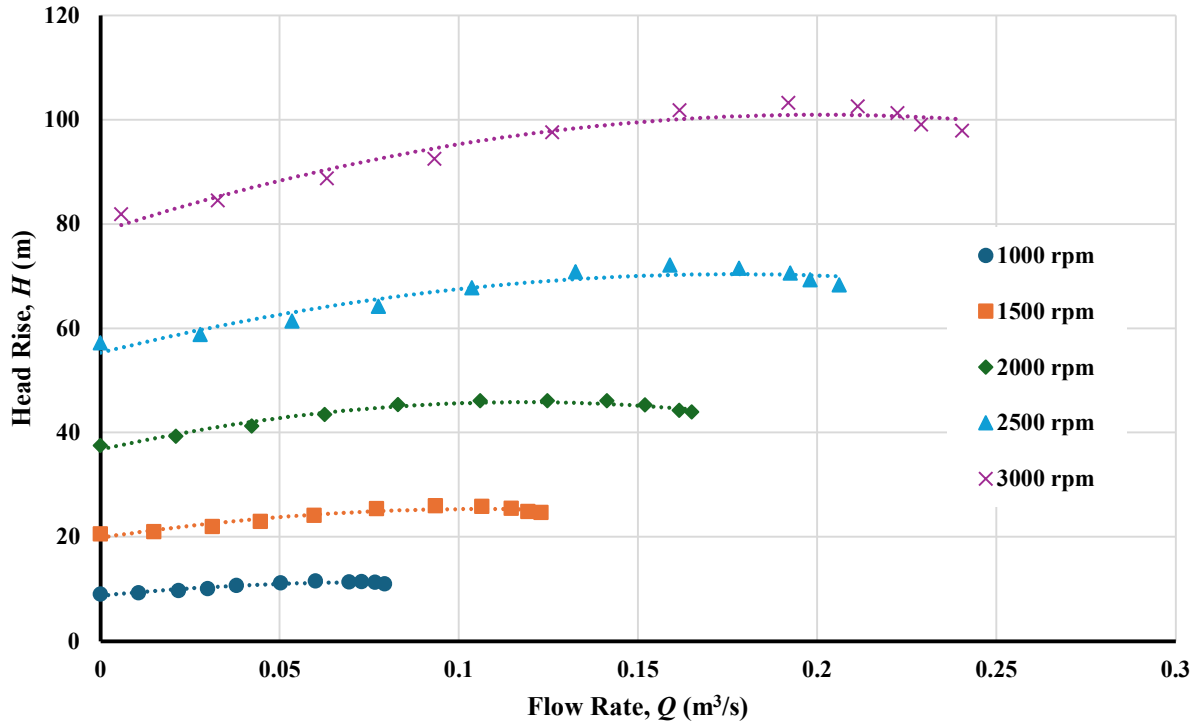

Figure 1: This graph showcases the relationship between head rise, *H*, and the volumetric flow rate of air leaving the nozzle, *Q*, for all five speeds tested on the radial blades impeller fan. Flow rate was calculated using the provided discharge coefficient, nozzle throat area, pressure differential across the nozzle and the air density. Head rise was found by taking the pressure differential across the fan and dividing by the product of air density and the gravitational constant. For all five speeds, eleven data points are plotted corresponding to each 10% increment as the slider plate was adjusted from 100% to 0%. As can be observed above, head rise and flow rate both increase proportionally with impeller speed. For each 500 rpm jump in speed, head rise increases by about 20 m and flow rate increases by approximately 0.05 m^3/s. The increase in head rise and volumetric flow rate with rotational speed follows the turbomachinery affinity laws, where flow rate scales linearly with speed and head rise scales with the square of speed due to the quadratic dependence of kinetic energy on blade tip velocity. However, around the seventh data point for each speed, the head rise peaks and then begins to slightly decrease while flow rate continues to increase. This is because higher volumetric flow produces greater internal losses and reduces the static pressure rise that the impeller can generate.

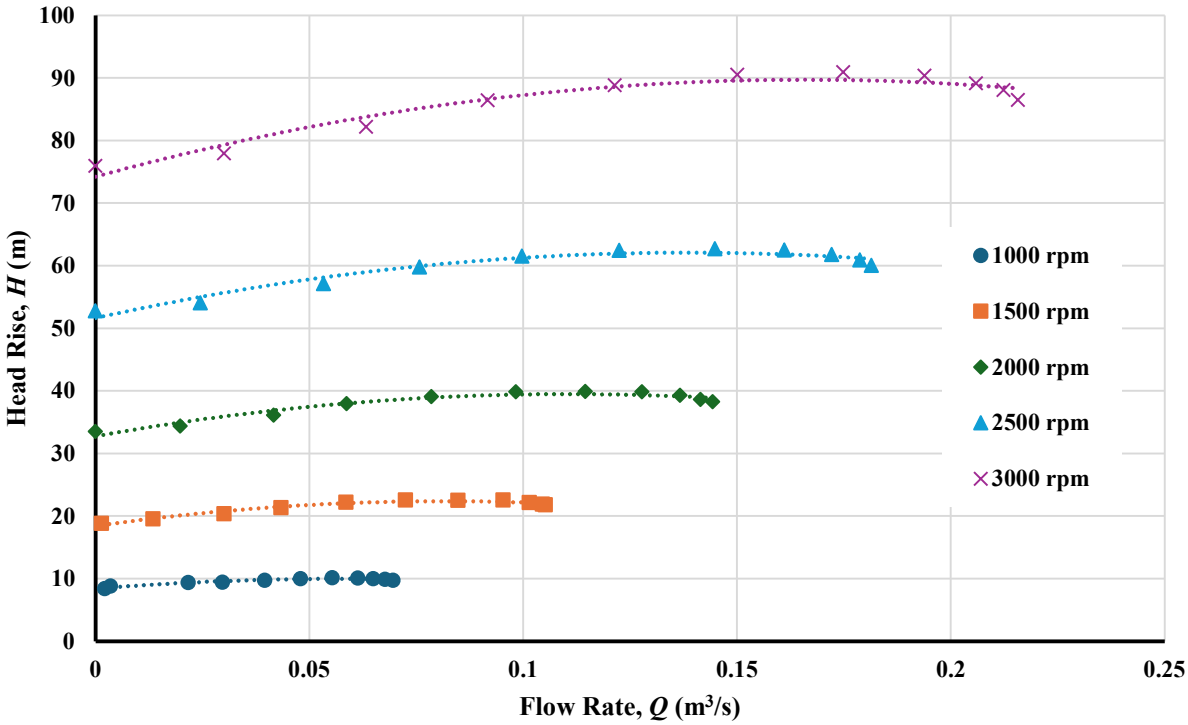

Figure 2: This graph displays the same relationship between head rise, *H*, and volumetric flow rate, *Q*, as that in Figure 1, but now for all five speeds tested on the backwards-inclined blades impeller fan. The same calculations were performed for each variable and nearly identical trends can be observed as before. The key difference for the backwards-inclined blades impeller fan is that the maximum flow rate and head rise values were consistently about 10% less than those values for the radial blades impeller fan. For example, the maximum head rise for the radial fan at 3000 rpm just exceeds 100 meters while the maximum head rise for the backwards-inclined fan at 3000 rpm was about 90 meters. The reason for these decreases is that the blade geometry of the backwards-inclined blades produces lower exit tangential velocity and therefore imparts less kinetic energy to the air than radial blades at the same rotational speed.

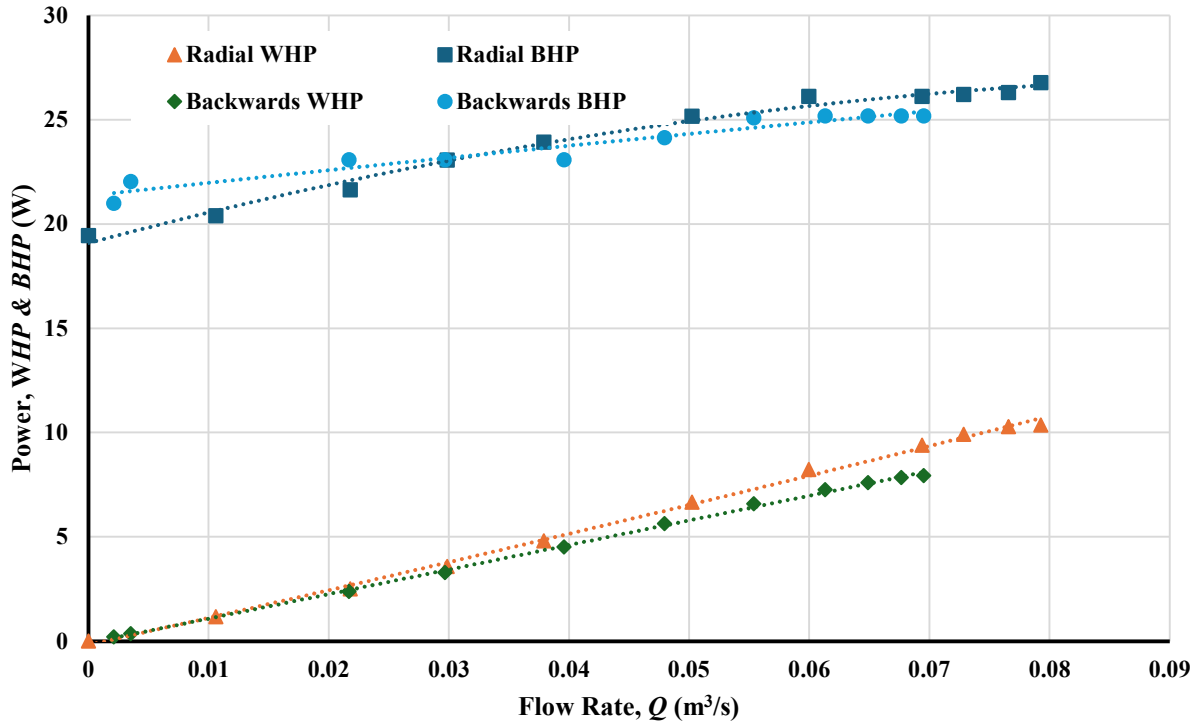

Figure 3: This graph shows the relationship of water horsepower, *WHP*, and brake horsepower, *BHP*, with volumetric flow rate, *Q,* for both the radial and backwards-inclined impeller fans at a speed of 1000 rpm. Water horsepower was calculated as the product of density, flow rate, head rise and the gravitational constant, brake horsepower was calculated as the product of the measured torque and impeller speed, and the same volumetric flow as the previous graphs was used. For both impeller types, the water horsepower began around 0 watts at the 0% slider plate position and increased linearly with flow rate. However, the radial fan achieved a greater maximum power and flow rate at 10 watts and 0.08 m^3/s, respectively, while the backwards-inclined fan lagged behind with maximum values of 8 watts and 0.07 m^3/s, respectively. For brake horsepower, a linearly increasing relationship with flow rate was also observed, but the starting values for power were much higher. While the water horsepower for each fan began at 0 watts due to its reliance on flow rate, the brake horsepower for the radial fan began at 20 watts and the backwards-inclined fan began around 21 watts. Despite the backwards-inclined fan initially having greater break horsepower than the radial fan, the values converged at a flow rate of 0.03 m^3/s and then the radial fan brake horsepower was greater for the remaining flow rates. The approximately linear increase in both water and brake horsepower with flow rate occurs because, at constant rotational speed, the mass flow rate through the fan increases proportionally with volumetric flow, resulting in a proportional increase in energy transfer to the fluid and aerodynamic loading on the impeller blades. The radial impeller exhibits overall greater horsepower because its blade geometry produces greater exit tangential velocity and pressure rise at the same speed, requiring greater torque input and transferring more energy to the air than the backwards-inclined design.

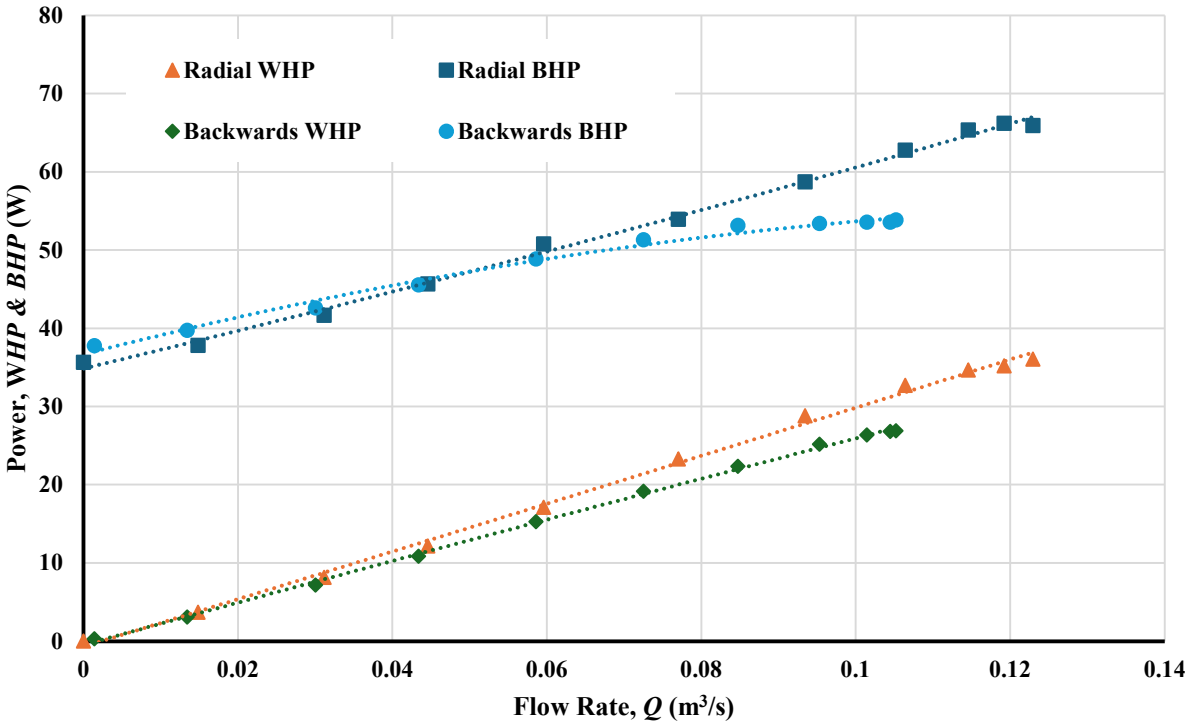

Figure 4: This graph displays the same relationship between the two horsepower metrics and volumetric flow rate for both fan types as Figure 3, but now for the 1500 rpm impeller speed data. The same exact trends can be observed, with the only difference being the magnitude of the values. Due to the greater impeller speed, flow rate values reached a maximum of roughly 0.125 m^3/s now compared to 0.08 m^3/s from before, and brake horsepower exceeded 65 watts and 52 watts for the radial and backwards-inclined impellers, respectively, whereas before they maxed out around 25 watts. The reason for the higher power values is that the increased rotational speed raised the impeller tip velocity which increased the momentum transfer to the fluid and therefore shifted the entire performance curve upward. The greater proportional increase in power than flow rate is consistent with turbomachinery affinity laws where flow scales with speed, head scales with speed squared, and power scales with speed cubed.

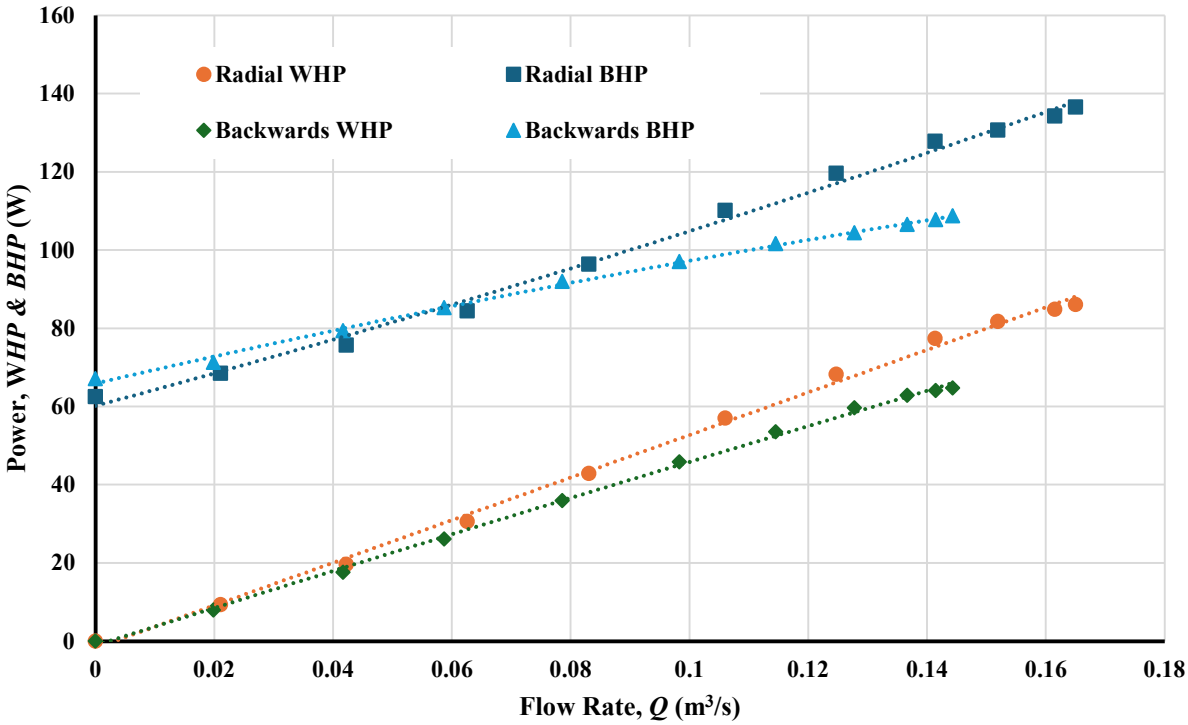

Figure 5: This graph is the same as the previous two, but now for an impeller speed of 2000 rpm for both fans. The same exact trends can be observed, with the only difference being the magnitude of the values. Due to the greater impeller speed, flow rate values reached a maximum of roughly 0.165 m^3/s now compared to 0.125 m^3/s from Figure 4, and brake horsepower reached nearly 140 watts and 65 watts for the radial and backwards-inclined impellers, respectively, whereas in Figure 4 they maxed out around 65 watts and 52 watts, respectively. The water horsepower for both fans increased proportionally with the brake horsepower as well, and the same reasoning as provideded for the increases in the data from Figure 3 to Figure 4 applies here.

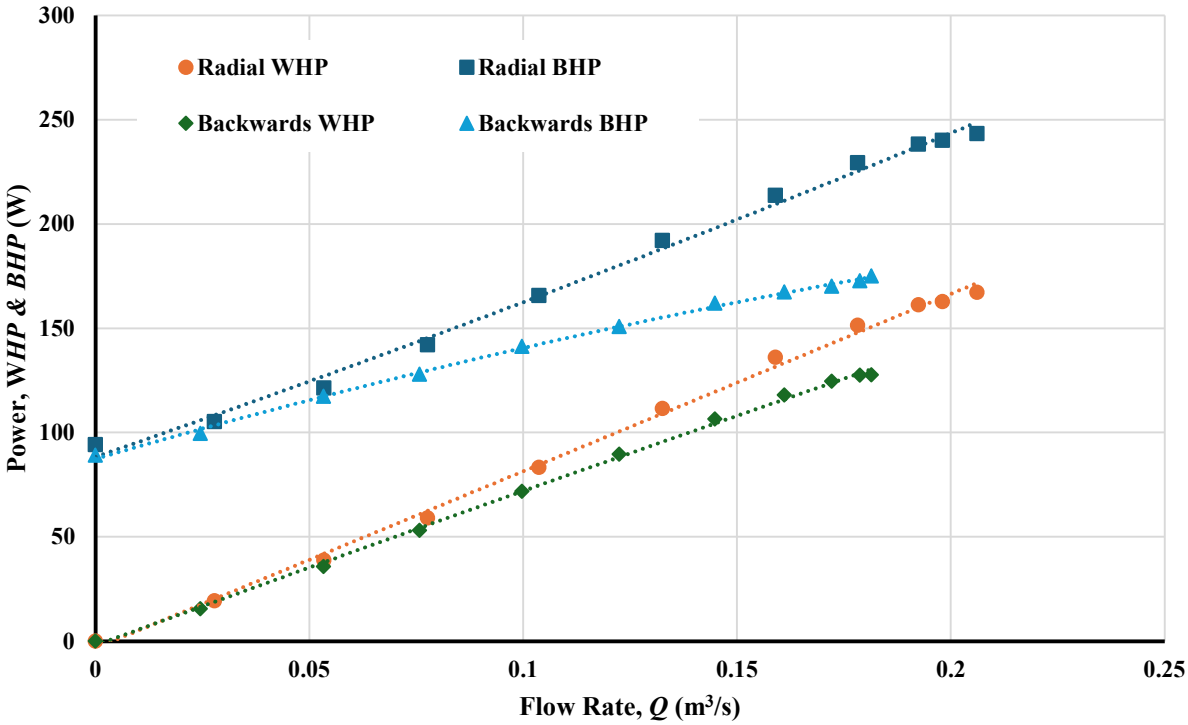

Figure 6: This graph is the same as the previous three, but now for an impeller speed of 2500 rpm for both fans. A different trend can now be observed for the brake horsepower data from the previous speeds. Whereas before the backwards-inclined fan brake horsepower began greater than the radial brake horsepower and remained greater until approximately the fourth data point, now the radial brake horsepower begins greater than the backwards-inclined horsepower from the first data point and remains greater for all flow rates tested. At higher rotational speeds, the radial impeller requires consistently greater torque due to its blade geometry imparting higher tangential velocity and pressure rise, which causes its brake horsepower to exceed that of the backwards-inclined impeller across the entire operating range. Additionally, the magnitudes of the water horsepower values for both fans approach much closer to the brake horsepower values than previously observed, especially the radial fan water horsepower which nearly exceeds the backwards-inclined fan brake horsepower. This is because as speed increases the fluid power rises more rapidly relative to mechanical losses, causing the water horsepower values to approach the brake horsepower values indicating improved overall efficiency at these higher operating conditions.

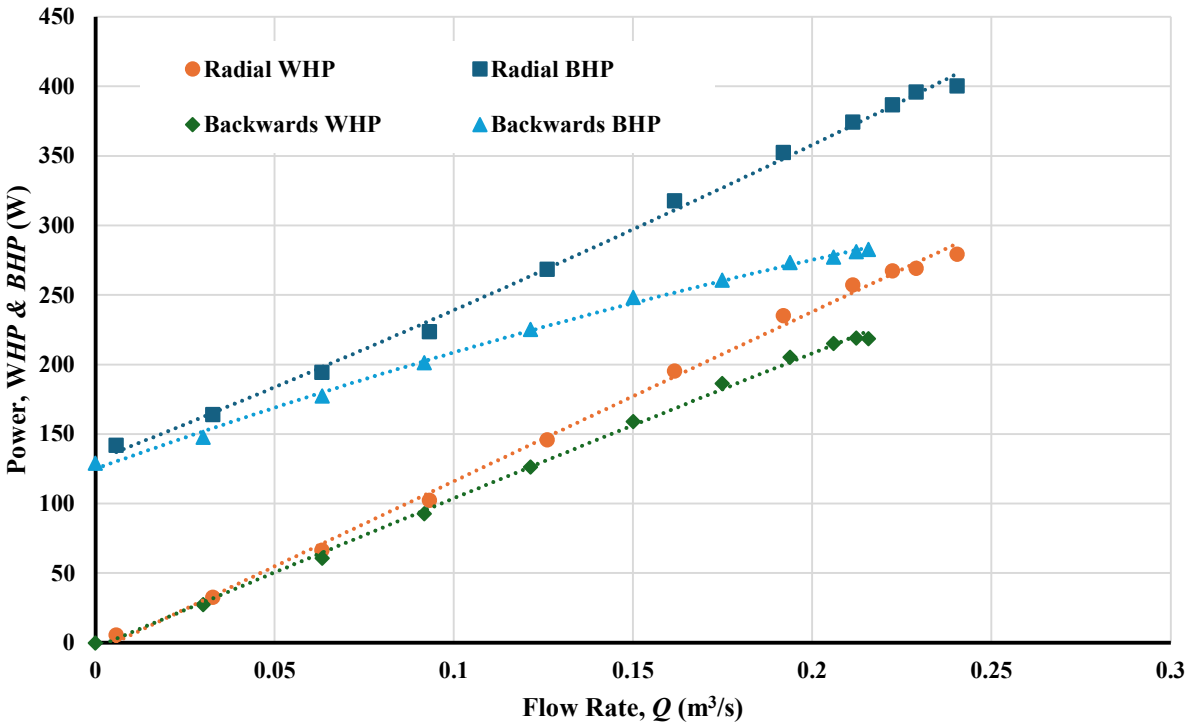

Figure 7: This graph is the same as the previous four, but now for an impeller speed of 3000 rpm for both fans. The same exact trends can be observed here, as those from Figure 6 for the 2500 rpm fan speed. However, due to the greater impeller speed, flow rate values reached a maximum of roughly 0.24 m^3/s now compared to 0.21 m^3/s from before, and brake horsepower exceeded 400 watts and 275 watts for the radial and backwards-inclined impellers, respectively, whereas before in Figure 6 they maxed out around 250 watts and 175 watts, respectively. The increase in power and flow rate with speed can once again be attributed to the greater impeller tip velocity and momentum transfer to the fluid.

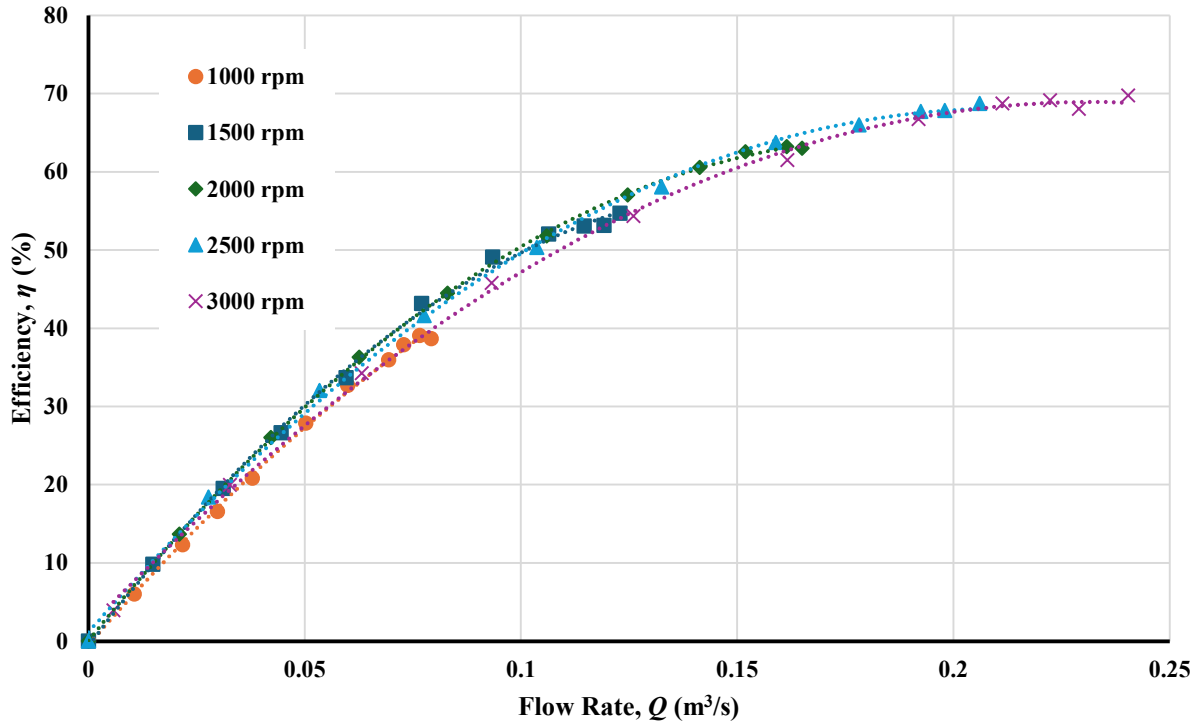

Figure 8: This graph showcases the efficiency curve for the radial blades impeller fan operating at all five speeds. The fan efficiency, η, was found by dividing water horsepower, *WHP*, by brake horsepower, *BHP*, and multiplying that quotient by 100 to express the ratio as a percentage. Volumetric flow rate, Q, was determined using the same method as previously discussed. The data points for all fan speeds follow the trajectory of the same curve because the fan performance is governed by similarity principles, causing dynamically similar operating points to collapse onto a common efficiency profile. However, a higher efficiency is achieved for each speed increase, with maximum efficiencies at speeds of 1000 rpm, 1500 rpm, 2000 rpm, 2500 rpm and 3000 rpm approximately equal to 40%, 55%, 63%, 69% and 70%, respectively. Each subsequent increase in speed experienced a proportional smaller increase in maximum fan efficiency, ranging from a 15% jump between 1000 rpm and 1500 rpm and a 1% jump between 2500 rpm and 3000 rpm. The rate of improvement decreases at higher speeds because work horsepower approaches brake horsepower asymptotically, meaning additional increases in rotational speed yield diminishing reductions in relative losses and therefore smaller gains in peak efficiency. Judging by the shape of the cure, peak efficiency for the radial fan will likely stay converged at 70% regardless of the impeller speed.

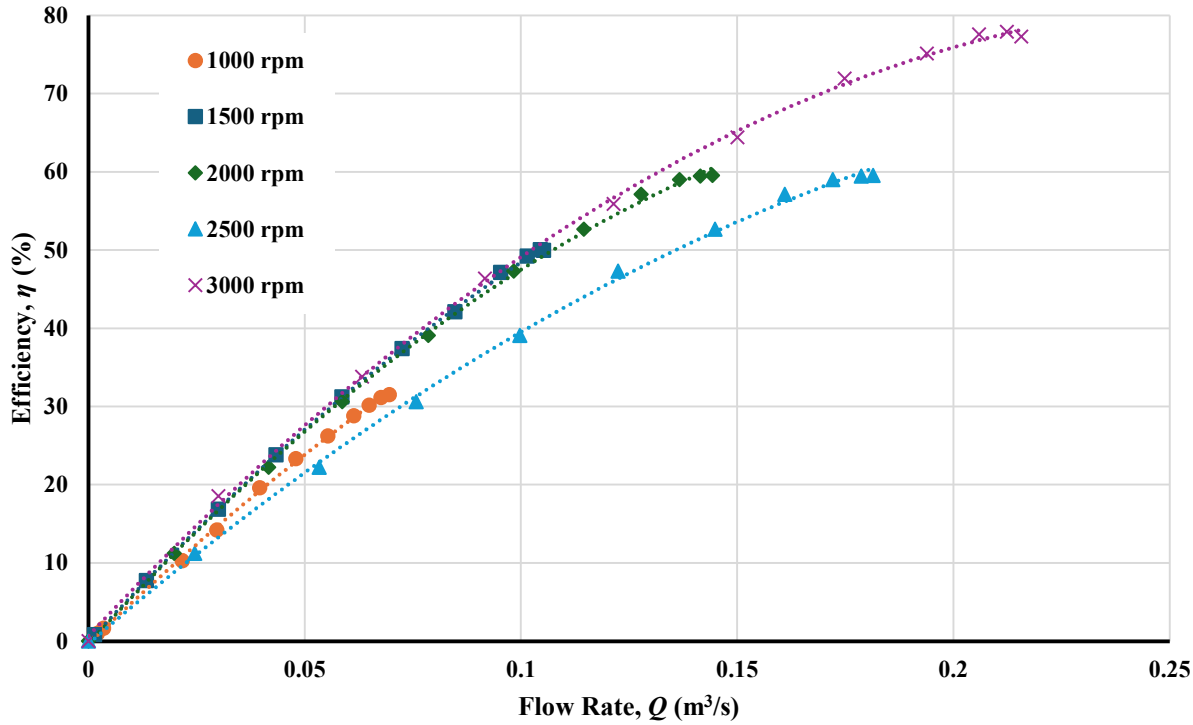

Figure 9: This graph showcases the efficiency curve for the backwards-inclined blades impeller fan operating at all five speeds. The fan efficiency, η, and volumetric flow rate, Q, were found using the same methods as before. While the data points for all speeds relatively follow the same trajectory, they are not as packed together as was observed for the radial fan in Figure 8, especially with the 2500 rpm data set serving as the biggest outlier. A potential cause is that the backwards-inclined impeller is more sensitive to changes in Reynolds number and blade loading across speeds, which can modify internal flow separation and loss mechanisms, causing the efficiency curves to deviate slightly from ideal similarity behavior. Another observable difference is that each speed for the backwards-inclined fan achieved a higher maximum efficiency with the 3000 rpm speed achieving nearly 80% whereas the same data set for the radial fan only reached 70%. The reason that the backwards-inclined impeller achieves a higher maximum efficiency is that its blade geometry reduces kinetic energy losses and flow separation, resulting in lower aerodynamic losses and more effective conversion of shaft power into useful pressure rise. Additionally, unlike the radial fan, the efficiency values for the backwards-inclined fan have not yet fully converged and therefore further increasing the rotational speed will achieve even greater efficiency, forecasted to reach about 85% based on the above trajectory.

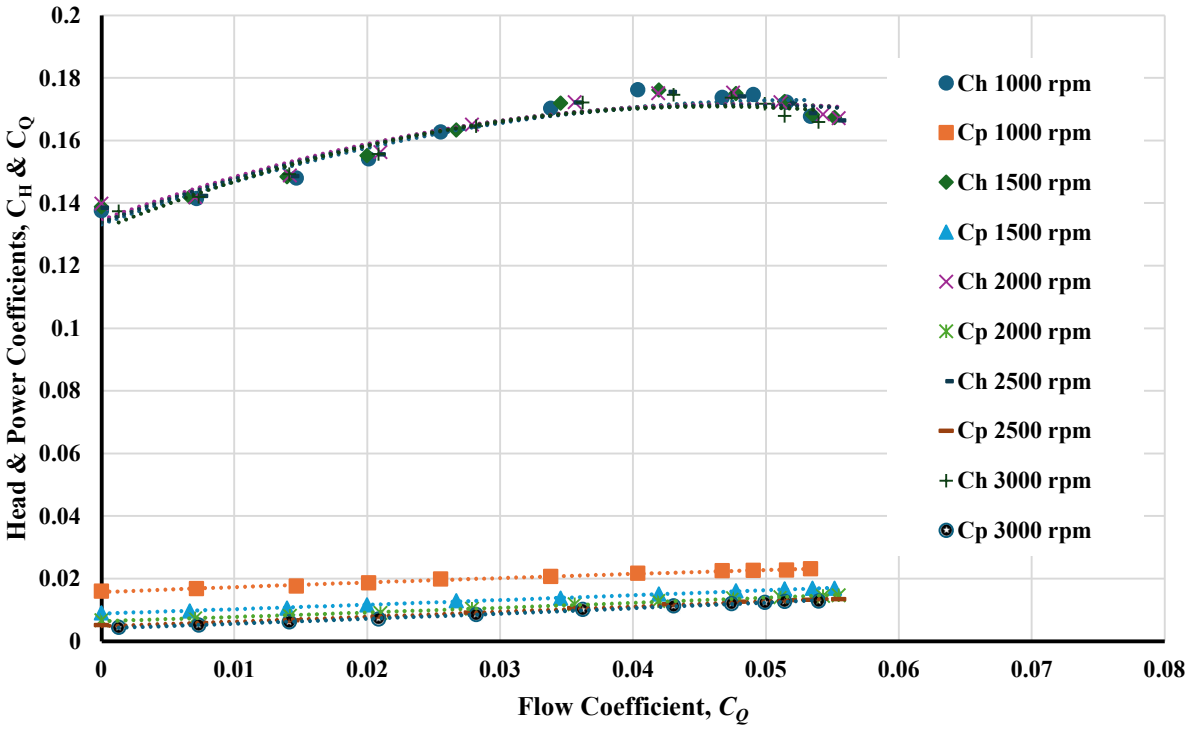

Figure 10: This graph shows the head coefficient, C_H, and the power coefficient, C_P, plotted against the flow coefficient, C_Q, for the radial blades impeller fan at all five speeds. The head coefficient was calculated by dividing the product of the gravitational constant and head rise by the product of the impeller speed (converted to radians per second) squared and the impeller diameter squared. Next, the power coefficient was determined by dividing brake horsepower by the product of air density, impeller speed cubed and impeller diameter to the fifth. Finally, the flow coefficient was found by dividing the volumetric flow rate by the product of the impeller speed and the impeller diameter cubed. All three of these coefficients are dimensionless and several trends can be observed from the graph. First off, both head and power coefficients linearly increased with an increasing flow coefficient, however, head coefficients displayed some non-linear behavior at higher flow rates due to internal losses. Next, for all impeller speeds, the power coefficient values were less than the head coefficient values and the higher impeller speeds produced lower power coefficients due to the inverse relationship of the two variables. On the other hand, the head coefficients did not vary among the different impeller speeds and all followed the same general curve positioned above all the power coefficient values.

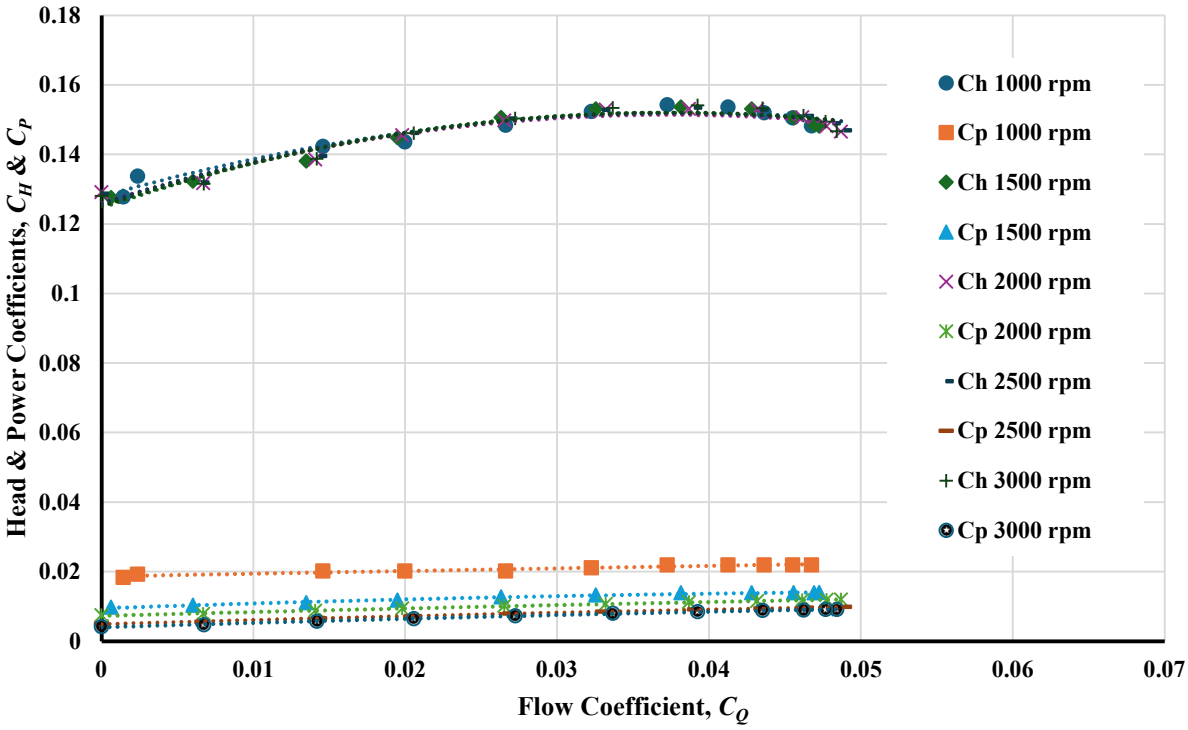

Figure 11: This graph shows the head coefficient, C_H, and the power coefficient, C_P, plotted against the flow coefficient, C_Q, for the backwards-inclined blades impeller fan at all five speeds. All coefficients were calculated in the same way as described for the radial fan and the same trends can be observed here as those in Figure 10. However, the values for head and flow coefficients are slightly lower than those observed in Figure 10. For example, for the radial fan, the maximum head and flow coefficient values were about 0.18 and 0.056, respectively, while for the backwards-inclined fan, they were about 0.16 and 0.05, respectively. The reason for these lower values is that the backwards-inclined blade geometry produces lower exit tangential velocity and therefore imparts less specific energy to the fluid at the same nondimensional operating condition. Since the coefficients scale energy transfer and flow relative to impeller speed and diameter, the reduced momentum exchange for the backwards-inclined design results in uniformly smaller nondimensional performance values compared to the radial impeller.

Table 1: This table displays both the average operating power, *P*, and the corresponding annual operating cost, *C*, of the backwards-inclined fan when the slide valve is fully open for each of the speeds, *N*, that it operated at. To calculate the annual cost, a time of 2800 operating hours was assumed per year as well as an electricity rate of 0.15 kWh. The motor was also assumed to have an efficiency of 80%. As power consumption and the fan speed increase, so does operating costs because when the fan spins faster it needs to move more air over smaller intervals of time. This would require more power, which requires more electricity and in turn, increases the annual operating cost. This is also in accordance with the fan affinity laws with flowrate increasing proportionally with speed, pressure increasing with the square of speed, and power increasing with the cube of speed.

Fan Speed, *N* (rpm)	**Power, *P* (W)**	**Annual Operating Cost, *C***
1000	26.00	$13.65
1500	54.54	$28.64
2000	109.00	$57.23
2500	174.81	$91.77
3000	283.00	$148.58

Table 2a: The following table portrays the fan performance changes for the radial fan at a speed of 1000 rpm and a corresponding mechanical power loss of 4 watts. Data is recorded when the slide valve is initially 100% open until it is fully closed at a slider plate position of 0%. The recorded data includes torque, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η. As the valve closes, the volumetric flowrate steadily decreases and eventually reaches a value of 0 m^3/s. This makes sense because as the valve slowly closes, airflow becomes increasingly limited. Furthermore, torque and power both gradually decrease due to their correlated relationship with flow rate whereas there is more flow, more work is required so higher torque and power are needed. The pressure downstream of the fan, P_3, gradually increases which represents a greater flow resistance since there is less space for the air to move through. Overall, the fan seems to operate most efficiently when the valve position is between 90% and 100% with high flow conditions and then worsens as the slide valve is closed.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
1001.0	100	0.2555	26.54	-211.45	-118.36	12.27	11.00	0.0793	38.68
1001.0	90	0.2509	26.09	-197.45	-113.82	20.36	11.29	0.0766	39.08
1001.0	80	0.2500	26.09	-178.73	-107.00	29.18	11.46	0.0729	37.87
1001.0	70	0.2491	26.00	-162.00	-93.18	42.18	11.39	0.0694	35.97
1001.0	60	0.2400	26.00	-120.91	-73.09	63.36	11.56	0.0599	32.71
1001.0	50	0.2282	24.81	-84.90	-53.45	79.18	11.16	0.0502	27.87
1001.0	40	0.2200	23.63	-48.36	-32.81	94.00	10.67	0.0379	20.84
1001.0	30	0.2064	21.63	-30.00	-19.90	100.18	10.11	0.0299	16.59
1001.0	20	0.1945	20.45	-16.00	-11.54	103.82	9.71	0.0218	12.33
1001.0	10	0.1855	19.54	-3.81	-1.81	108.36	9.27	0.0106	06.02
1001.0	0	0.1764	18.63	0.00	5.90	113.09	9.02	0.00	0.00

Table 2b: This table portrays experimental and calculated data from the radial fan operating at a speed of 1500 rpm with a mechanical power loss of 5 W. The system torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η, is highlighted. This fan produced a higher head rise and flowrate indicating that there is a large pressure increase and a larger volume of air moving through the system. As the slide valve gradually closes towards being fully shut, flow rate steadily decreases with an increase in downstream pressure due to flow resistance. The fan is again more efficient as the slide valve is more open. However, a sharper decrease in efficiency arises which indicates that as the fan speed increases, the overall system may lose efficiency more easily as the valve is closed. This happens because as the fan spins faster, more energy is applied to the air. The air then gets restricted and more energy is wasted, so the flow is blocked leading to a less efficient system.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
1501.6	100	0.4191	65.909	-508.55	-266.82	26.546	24.700	0.1229	54.72
1502.0	90	0.4209	66.183	-478.55	-253.00	42.182	24.847	0.1192	53.16
1502.0	80	0.4155	65.546	-442.00	-236.36	66.181	25.467	0.1146	53.06
1502.0	70	0.3991	62.909	-381.27	-207.00	100.09	25.849	0.1064	52.06
1500.6	60	0.3736	59.364	-293.82	-162.82	145.73	25.972	0.0934	49.10
1503.0	50	0.3427	55.273	-199.82	-114.55	187.64	25.436	0.0770	43.16
1503.0	40	0.3227	50.636	-119.45	-70.364	216.64	24.158	0.0596	33.67
1503.6	30	0.2900	46.000	-66.909	-42.818	229.91	22.961	0.0449	26.63
1504.0	20	0.2645	42.455	-32.727	-22.636	238.45	21.983	0.0312	19.54
1504.0	10	0.2400	39.000	-7.4545	-2.7273	247.09	21.028	0.0149	09.81
1504.0	0	0.2264	36.636	0.00	11.727	255.91	20.558	0.00	0.00

Table 2c: This table portrays the recorded data for the radial fan operating at a speed of 2000 rpm with a mechanical power loss of 9 W. The system torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η, are recorded. The total head rise and volumetric flowrates are much larger, indicating a greater energy transfer to the air at this specific condition. The decreasing torque indicates that the fan is doing less work on the air as the slide valve closes. Since a smaller amount of air is being moved, the motor does not need to supply as much rotational force to the system, hence lowering the total torque. The fan is over 60% efficient when the slide valve is fully open validating the fact that as motor speed increases, the fan becomes more efficient. When the fan spins faster, it can move air much more smoothly which allows more energy from the motor to be applied to the moving air making the system more efficient overall. It too becomes completely inefficient when the slide valve is fully closed which makes sense because flow rate is 0. The negative pressures also indicate a sub-atmospheric value which is what is supposed to happen at the inlet when air is pulled in due to the introduction of suction.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
2004.0	100	0.6509	137.09	-915.82	-473.00	48.63	43.93	0.1649	63.00
2004.0	90	0.6400	135.00	-877.27	-450.09	75.63	44.26	0.1615	63.20
2004.0	80	0.6227	131.55	-776.91	-401.18	136.91	45.29	0.1519	62.55
2004.0	70	0.6091	128.82	-672.36	-350.27	197.27	46.09	0.1413	60.54
2005.0	60	0.5700	119.00	-523.27	-279.45	268.18	46.09	0.1247	57.05
2005.3	50	0.5245	110.91	-378.00	-205.82	332.64	45.34	0.1059	51.81
2006.0	40	0.4591	97.81	-232.00	-132.36	384.18	43.49	0.0830	44.47
2007.6	30	0.4018	85.18	-131.82	-78.81	410.91	41.24	0.0626	36.29
2008.0	20	0.3600	77.00	-60.00	-38.63	428.09	39.29	0.0422	26.03
2008.0	10	0.3255	68.09	-14.90	-3.18	442.00	37.48	0.0210	13.68
2009.0	0	0.2973	62.45	0.00	23.72	462.45	36.94	0.00	0.00

Table 2d: This table portrays the data for the radial fan at an operating speed of 2500 rpm and a mechanical power loss of 12 W. The system torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η, is taken as data. With the fan operating at a very high speed, the head rise and volumetric flowrate are much higher. The head rise is roughly 50-60 m more through all the data points than when the fan was operating at 1000 rpm. The torque and total power in the system additionally decrease with the closing of the slide valve due to a smaller amount of air being moved. The total power is larger overall than it would be with a lower fan speed because of a greater energy input. Additionally, the fan maintains a relatively high efficiency until the slide valve is right about to be closed. That constant large efficiency value explains that the fan is still generating a lot of pressure even with a decrease in flow. It is spinning so fast that there is enough work to keep the system efficient even when the valve is closing.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
2504.0	100	0.9282	245.18	-1429.84	-737.82	73.72	68.32	0.2061	68.72
2504.1	90	0.9155	242.09	-1319.36	-699.00	123.64	69.27	0.1979	67.85
2504.8	80	0.9082	237.64	-1245.83	-642.00	196.00	70.58	0.1924	67.69
2504.9	70	0.8745	229.91	-1069.12	-533.55	296.09	71.51	0.1782	66.00
2505.6	60	0.8145	215.45	-850.36	-440.82	416.09	72.13	0.1589	63.72
2507.0	50	0.7318	193.36	-591.09	-311.36	530.18	70.84	0.1325	58.04
2508.5	40	0.6309	166.27	-361.64	-196.91	608.00	67.75	0.1036	50.34
2510.1	30	0.5409	142.45	-202.73	-115.45	647.00	64.17	0.0776	41.63
2511.7	20	0.4618	123.36	-96.00	-58.27	670.82	61.39	0.0534	32.06
2512.5	10	0.4000	106.00	-26.00	-6.45	691.91	58.78	0.0278	18.44
2512.2	0	0.3582	95.63	0.00	40.00	719.64	57.21	0.00	0.00

Table 2e: This table includes data for the radial fan at its fastest operating speed of 3000 rpm with a mechanical power loss of 14 W. The torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η, is recorded. The head rise and volumetric flow rate reach their highest values, which is consistent with the trend of the data as fan speed increases. The fact that torque and total power are at their highest values also indicates how much more energy is needed at this operating condition. The fan is extremely efficient throughout most of the run time no matter where the slide valve is. It is even slightly efficient when the slide valve is fully closed which means that there is still a small leakage of airflow allowing the fan to function with an essentially closed valve, however, the efficiency should be 0% in an ideal case. Overall, the fan seems to operate best with the highest speeds.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
3001.7	100	1.27	402.36	-1946.00	-1507.49	105.36	97.87	0.2404	69.77
3001.7	90	1.26	398.73	-1765.36	-1002.68	174.00	99.04	0.2289	68.02
3001.5	80	1.23	387.27	-1664.48	-901.36	301.55	101.27	0.2224	69.13
3003.3	70	1.19	376.18	-1502.96	-792.64	426.00	102.58	0.2113	68.74
3004.4	60	1.12	353.18	-1240.06	-633.18	592.82	103.19	0.1919	66.72
3006.3	50	1.01	318.00	-879.09	-452.27	757.64	101.84	0.1616	61.49
3010.0	40	0.85	269.82	-534.36	-285.27	874.27	97.61	0.1259	54.35
3013.1	30	0.70	225.72	-292.55	-164.36	934.73	92.51	0.0932	45.79
3013.9	20	0.61	195.90	-134.55	-81.45	972.73	88.73	0.0632	34.26
3015.6	10	0.52	165.00	-36.00	-5.54	998.36	84.50	0.0327	19.99
3016.7	0	0.45	145.00	-1.81	57.81	1030.59	81.87	0.0058	3.95

Table 3a: This data portrays the resulting conditions for the backwards impeller at its lowest operating fan speed of 1000 rpm and mechanical power loss of 4 W. The data recorded for this fan includes torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η. The head rise and fan efficiency are lower at this operating speed than they were for the radial fan likely due to the geometry of the blades. Backwards-inclined impeller fans are used to reduce power consumption and operate smoothly while limiting the amount of energy transferred to the air. Therefore, at lower speeds less pressure is generated meaning there is less fluid power and a less efficient system. The torque is quite consistent but does decrease with a closing valve position towards the end since the fan requires less rotational force there. The pressure drop also decreases due to less need for a vacuum as the valve closes. Overall, this system proves its efficiency and verifies what should happen as the slide valve closes.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
1002.0	100	0.24	26.00	-160.54	-100.27	13.90	9.74	0.0695	31.52
1002.0	90	0.24	26.00	-152.00	-97.81	18.09	9.89	0.0676	31.14
1002.0	80	0.24	26.00	-139.81	-92.09	25.00	9.99	0.0648	30.17
1002.0	70	0.24	26.00	-124.90	-81.36	37.00	10.10	0.0613	28.82
1002.0	60	0.23	25.90	-101.81	-68.81	50.00	10.13	0.0553	26.23
1002.0	50	0.23	25.00	-76.36	-52.00	65.36	10.01	0.0479	23.32
1002.0	40	0.22	24.00	-52.00	-35.27	79.00	9.75	0.0395	19.58
1002.0	30	0.22	24.00	-29.27	-20.81	89.81	9.44	0.0296	14.22
1002.0	20	0.22	23.00	-15.63	-11.45	98.18	9.35	0.0216	10.30
1002.0	10	0.21	22.00	-0.9091	-1.45	101.54	8.78	0.0035	1.64
1002.0	0	0.20	21.00	0.5455	5.72	104.18	8.40	0.0021	0.99

Table 3b: This table shows the recorded data for the backwards-inclined fan at an operating speed of 1500 rpm with a mechanical power loss of 5 W. Both experimental and calculated data are recorded for this fan including torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η. There seems to be an increase in torque which makes sense due to the increase in fan speed. An interesting point to note is that the torque when the slide valve is fully closed is the same as the torque when the slide valve is fully open at an operating speed of 1000 rpm. Although there may be different loads on the blade at different speeds, the total resistance on those blades may be the same depending on the position of the valve. The fan is nearly halfway to maximum efficiency when the valve is fully open but also has less power output than a radial fan might have. The fan even had slight efficiency when the valve was fully closed but this could have been due to error in closing the valve. Additionally, this data indicates that the backwards-inclined fan allows for safer operation and prevents motor overloading during restrictions.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
1500.0	100	0.3427	54.54	-367.45	-218.72	37.00	21.82	0.1051	49.97
1500.0	90	0.3409	54.18	-362.36	-217.45	39.00	21.88	0.1044	50.03
1500.0	80	0.3409	54.18	-341.63	-202.27	57.54	22.17	0.1014	49.21
1500.0	70	0.3400	53.90	-301.63	-182.63	81.54	22.54	0.0953	47.14
1496.2	60	0.3391	52.81	-238.36	-143.72	120.18	22.52	0.0847	42.08
1501.0	50	0.3264	51.27	-174.54	-110.81	153.72	22.57	0.0725	37.39
1501.0	40	0.3109	49.09	-114.00	-75.00	185.36	22.21	0.0585	31.21
1500.0	30	0.2900	46.00	-62.54	-40.63	209.36	21.33	0.0433	23.81
1501.0	20	0.2709	43.09	-30.00	-20.27	218.45	20.36	0.0300	16.85
1502.0	10	0.2527	40.27	-6.00	-4.36	224.63	19.54	0.0134	7.74
1502.0	0	0.2400	39.00	0.3636	12.45	233.36	18.84	0.0014	0.82

Table 3c: This table portrays data for the backwards-inclined fan at a middle range operating speed of 2000 rpm with a mechanical power loss of 9 W. This data includes both experimental and calculated values including torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η. The fan performs quite well at this operating condition, with an efficiency well over 50% up until the slide valve is almost halfway closed. The increase in overall head rise and volumetric flow rate is consistent with the relationship increasing fan speed. Again, total torque and power gradually begin to decrease proving that the backwards fan avoids motor overloads at low volumetric flowrates. This data displays a more controlled behavior and higher speeds compared to lower operating conditions with a fan efficiency of 0% when the slide valve is fully closed as expected. Any offset variables may be from the fact that the fan speed is not precisely 2000 rpm despite being assumed to be at a constant speed of 2000 rpm.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
1997.0	100	0.5200	109.00	-691.63	-397.09	51.36	38.27	0.1443	59.51
1996.3	90	0.5155	108.09	-664.36	-380.54	72.27	38.64	0.1414	59.44
1995.3	80	0.5100	107.00	-620.00	-354.18	106.09	39.27	0.1366	59.01
1995.0	70	0.5000	105.00	-542.18	-313.00	153.90	39.84	0.1277	57.11
1995.0	60	0.4864	102.27	-435.45	-253.63	213.72	39.88	0.1145	52.68
1995.0	50	0.4645	98.45	-320.90	-190.54	276.18	39.83	0.0983	47.28
1996.0	40	0.4400	92.00	-204.90	-125.18	332.54	39.06	0.0785	39.09
1996.0	30	0.4082	85.63	-114.36	-73.81	371.09	37.96	0.0586	30.60
1996.0	20	0.3800	80.00	-57.63	-38.36	385.09	36.13	0.0416	22.20
1996.0	10	0.3409	73.09	-13.09	-8.72	394.09	34.37	0.0198	11.21
1991.2	0	0.3218	67.18	0.00	19.18	412.00	33.52	0.00	0.00

Table 3d: This table shows data for the backwards-inclined fan at a high operating speed of 2500 rpm with a mechanical power loss of 12 W. The data includes both experimental and calculated values including torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η, for the system. Efficiency values spiked at this much higher speed. Volumetric flow rate steadily decreases with a closing slide valve in addition to the pressure drop, indicating a greater system resistance. At this speed, the fan is more than twice as efficient when the sliding valve is mostly open compared to what the fan was at during its lowest operating speed of 1000 rpm for the same flow condition. The higher head rise portrays a stronger pressure-building capability. The fact that the head rise essentially doubled from when the fan was at 2000 rpm explains that it does not take much for more energy to be transferred to the air. Overall, the increase in fan speed enhances the pressure and flow performance significantly, with the backwards design maintaining a higher efficiency and more stable power behavior.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
2499.0	100	0.6691	174.81	-1092.96	-624.81	78.90	60.05	0.1814	72.91
2499.0	90	0.6600	173.00	-1060.32	-60 2.27	111.45	60.89	0.1786	73.83
2499.0	80	0.6500	170.00	-983.63	-555.90	167.81	61.76	0.1721	73.23
2499.0	70	0.6400	168.00	-861.27	-488.90	243.90	62.52	0.1610	70.46
2499.0	60	0.6191	162.81	-696.90	-400.27	334.90	62.72	0.1448	65.73
2499.0	50	0.5764	152.90	-497.81	-288.09	444.09	62.47	0.1224	59.43
2500.0	40	0.5400	142.00	-330.18	-195.63	525.27	61.52	0.0997	50.85
2500.0	30	0.4891	129.72	-190.54	-118.72	582.00	59.79	0.0757	41.45
2501.1	20	0.4482	117.45	-94.54	-62.90	607.00	57.14	0.0533	30.44
2502.0	10	0.3800	101.00	-20.00	-14.09	619.45	54.06	0.0245	15.61
2503.0	0	0.3400	90.81	0.00	30.27	649.18	52.77	0.00	0.00

Table 3e: This data portrays results for the backwards-inclined fan at its highest operating speed of 3000 rpm with a mechanical power loss of 14 W. The data includes both experimental and calculated values from running the fan including torque, τ, total power, P, pressures ΔP_1, P_2, and P_3, head rise, H, volumetric flow rate, Q, and fan efficiency, η, for the system. As expected, the system is most efficient at the highest fan speed. It also seems to drop quite fast to 0% once the slide valve position is 20% open. The head rise and flowrate are both much higher overall than when the fan is at lower speeds. Pressure tends to build rapidly as the slide valve closes with a steady decrease in torque and power. This data reflects a stable motor under heavy loading and resistance. This data emphasizes how the backwards fan delivers a very high performance while keeping controlled and powerful behavior. The increased power indicates that the fan requires much more energy as it spins faster.

Fan Speed, N (rpm)	**Slide Valve Position (%)**	**Torque, τ (Nm)**	**Total Power, P (W)**	**ΔP_1 (Pa)**	**P_2 (Pa)**	**P_3 (Pa)**	**Head Rise, H (m)**	**Volumetric Flowrate, Q (m^3/s)**	**Fan Efficiency, η (%)**
3002.0	100	0.9000	283.00	-1545.00	-897.27	116.45	86.50	0.2156	77.27
3001.0	90	0.8955	281.63	-1497.26	-869.27	163.00	88.07	0.2123	77.88
3001.0	80	0.8827	277.81	-1408.57	-807.09	238.00	89.13	0.2059	77.57
3001.0	70	0.8700	274.00	-1248.35	-717.72	341.72	90.36	0.1938	75.11
3001.9	60	0.8300	263.00	-1015.83	-580.72	485.63	90.93	0.1748	71.45
3003.0	50	0.7900	248.00	-747.81	-431.63	629.27	90.52	0.1500	64.08
3004.0	40	0.7164	226.90	-489.63	-285.63	755.54	88.83	0.1214	56.10
3006.0	30	0.6400	202.00	-279.45	-169.09	844.54	86.44	0.0917	46.14
3007.6	20	0.5636	177.09	-132.90	-83.63	879.81	82.17	0.0632	34.33
3008.2	10	0.4691	149.72	-30.00	-20.18	893.63	77.94	0.0300	18.58
3010.0	0	0.4100	130.00	0.00	45.18	935.63	75.96	0.00	0.00

Sample Calculation of Density, ρ:

The following sample calculation is for the density of the air from the 1000 rpm and 100% slider position radial fan data set. The ideal gas law shown below was used to solve the air density, ρ, where P_{atm} is the atmospheric pressure of the room measured with a barometer, T is the ambient room temperature measured with a thermometer and R is the specific gas constant for air.

$$\rho = \frac{P_{atm}}{RT} \; Eqn.\,1.1, p.\,4 \; Fox \;\&\; McDonald \; [1] \tag{1}$$

The measured values for all the input variables of the equation are detailed below with all units converted to metric and temperatures converted to Kelvin where necessary. Then, the values were substituted into (1) to solve air density.

$$P_{atm} = 1025.456\, mbar * \frac{100\, Pa}{1\, mbar} = 102545.6\, Pa$$

$$T = 22°\mathrm{C} + 273.15 = 295.15\, K$$

$$R = 287.05\, \frac{J}{kgK} \; The\ Engineering\ ToolBox \; [2]$$

$$\rho = \frac{102545.6\, Pa}{287.05 \frac{J}{kgK} * 295.15\, K}$$

$$= \frac{102545.6\, \frac{kg\cancel{m}}{m^2\cancel{s^2}}}{287.05 \frac{\cancel{kg}m^2}{\cancel{kgKs^2}} * 295.15\, \cancel{K}}$$

$$\therefore \rho = 1.2104\, \frac{kg}{m^3}$$

Sample Calculation of Head Rise, *H*:

The sample calculation for head rise, *H*, of the air was completed for the radial fan speed of 1000 rpm and slide valve position of 100%. Head rise is essentially how much energy per unit weight the fan adds to the air to allow it to move through the system. Solving for head rise is dependent on the fluid pressure right before air enters the fan, P_2, the fluid pressure right after air exits the fan, P_3, air density, ρ, and gravity, g.

$$H = \frac{P_3 - P_2}{\rho g} \; p.6 \; Schulze \; [3] \tag{2}$$

All variables used in this calculation were in a metric unit system to stay consistent. This equation essentially takes the pressure increase across the distance of the fan and converts it to an equivalent fluid height to portray how much energy per unit weight is added to the air.

$$P_2 = -118.36 \; Pa$$

$$P_3 = 12.273 \; Pa$$

$$g = 9.81 \; \frac{m}{s^2}$$

$$\rho = 1.2104 \; \frac{kg}{m^3}$$

$$H = \frac{12.273 \; Pa - (-118.36 \; Pa)}{1.2104 \; \frac{kg}{m^3} * 9.81 \; \frac{m}{s^2}}$$

$$H = \frac{130.64 \; \frac{\cancel{kg}m}{\cancel{m^2}\cancel{s^2}}}{1.2104 \; \frac{\cancel{kg}}{\cancel{m^3}} * 9.81 \; \frac{\cancel{m}}{\cancel{s^2}}}$$

$$\therefore H = 11.002 \; m$$

Sample Calculation of Flow Rate, *Q*:

The following sample calculation is for the flow rate of air from the 1000 rpm and 100% slider position radial fan data set. The flow rate equation shown below was used to solve for the flow rate, Q, where C_D is the coefficient of discharge representing the difference between theoretical flow and actual flow, A is the cross-sectional area of the nozzle, ΔP_1 is the pressure drop at the nozzle, and ρ is the density of the air.

$$Q = C_D A \sqrt{\frac{2|\Delta P_1|}{\rho}} \; Eqn.1, p.26 \; TecQuipment\ Ltd \; [4] \tag{3}$$

The value measured for each input variable is shown below with units converted to metric. There were no temperatures present, so a conversion to Kelvin was unnecessary. The values were then substituted into (3) to solve for flow rate.

$$C_D = 10.356$$

$$A = 0.3869\ m^2$$

$$\Delta P_1 = -211.45\ Pa$$

$$\rho = 1.2104\ \frac{kg}{m^3}$$

$$Q = 10.356 * 0.3869\ m^2 \sqrt{\frac{2|-211.45\ Pa|}{1.2104\ \frac{kg}{m^3}}}$$

$$= 10.356 * 0.3869\ m^2 \sqrt{\frac{2\left|-211.45\ \frac{\cancel{kg}m}{\cancel{m^2}s^2}\right|}{1.2104\ \frac{\cancel{kg}}{\cancel{m^3}}}}$$

$$\therefore Q = 0.0793\ \frac{m^3}{s}$$

Sample Calculation of Water Horsepower, *WHP*:
The following sample calculation is for the water horsepower of air from the 1000 rpm and 100% slider position radial fan data set. The water horsepower equation shown below was used to solve for the water horsepower, WHP, where ρ is the density of the air, g is the acceleration due to gravity, Q is the flow rate through the nozzle, and H is the head rise of the air.

$$WHP = \rho g Q H \; p.6 \; Schulze \; [5] \tag{4}$$

The measured value for each input variable is shown below with units converted to metric. There were no temperatures present, so a conversion to Kelvin was unnecessary. The values were then substituted into (4) to solve for water horsepower.

$$\rho = 1.2104 \; \frac{kg}{m^3}$$

$$g = 9.81 \; \frac{m}{s^2}$$

$$Q = 0.0793 \; \frac{m^3}{s}$$

$$H = 11.002 \; m$$

$$WHP = 1.2104 \; \frac{kg}{\cancel{m^3}} * 9.81 \; \frac{m}{s^2} * 0.0793 \; \frac{\cancel{m^3}}{s} * 11.002 \; m$$

$$\therefore WHP = 10.356 \; W$$

Sample Calculation of Brake Horsepower, *BHP*:
The following sample calculation is for the brake horsepower from the 1000 rpm and 100% slider position radial fan data set. The brake horsepower equation shown below was used to solve for the brake horsepower, BHP, where τ is the torque and N is the revolutions per minute.

$$BHP = \tau * \left(\frac{2\pi N}{60}\right) \; p.6 \; Schulze \; [6] \tag{5}$$

The measured value for each input variable is shown below with units converted to metric. There were no temperatures present, so a conversion to Kelvin was unnecessary. The values were then substituted into (5) to solve for water horsepower. The revolutions per minute were converted in the equation to radians per second, rather than beforehand.

$$\tau = 0.2555 \; Nm$$

$$N = 1001.1 \; rpm$$

$$BHP = 0.2555 \; Nm * \left(\frac{2\pi \; \frac{\cancel{rad}}{\cancel{rev}} * 1001.1 \; \frac{\cancel{rev}}{\cancel{min}}}{60 \; \frac{s}{\cancel{min}}}\right)$$

$$\therefore BHP = 26.778 \; W$$

Sample Calculation of Efficiency, η:
The following sample calculation is for the efficiency of the pump from the 1000 rpm and 100% slider position radial fan data set. The ratio shown below was used to solve for the efficiency, η, where WHP is the water horsepower and BHP is the brake horsepower.

$$\eta = \frac{WHP}{BHP}\ p.7\ Schulze\ [5] \tag{6}$$

The measured value for each input variable is shown below with units converted to metric as necessary. There were no temperatures present, so a conversion to Kelvin was unnecessary. The values were then substituted into (6) to solve for efficiency.

$$WHP = 10.356\ W$$

$$BHP = 26.778\ W$$

$$\eta = \frac{10.356\ \cancel{W}}{26.778\ \cancel{W}}$$

$$\therefore \eta = 0.3869$$

Sample Calculation of Head Coefficient, C_H:
The following sample calculation is for the head coefficient of the fan from the 1000 rpm and 100% slider position radial fan data set. The head coefficient is a measure of the pressure increase, sometimes referred to as head. The equation shown below was used to solve for the head coefficient, C_H, where g is the acceleration due to gravity, H is the head rise of the air, N is the revolutions per minute, and D is the diameter of the fan impeller.

$$C_H = \frac{gH}{N^2 D^2} \; p.3 \; Schulze \; [7] \tag{7}$$

The measured value for each input variable is shown below with units converted to metric. There were no temperatures present, so a conversion to Kelvin was unnecessary. The revolutions per minute were converted to radians per second. The values were then substituted into (7) to solve for the head coefficient.

$$g = 9.81 \; \frac{m}{s^2}$$

$$H = 11.002 \; m$$

$$N = 1001.1 \cancel{rpm} * \frac{2\pi \; rad}{1 \cancel{rev}} * \frac{1 \; \cancel{min}}{60 \; s} = 104.82 \frac{rad}{s}$$

$$D = 0.242 \; m$$

$$C_H = \frac{(9.81 \; \frac{\cancel{m}}{\cancel{s^2}})(11.002 \; \cancel{m})}{(104.82 \frac{rad}{\cancel{s}})^2 (0.242 \; \cancel{m})^2}$$

$$\therefore C_H = 0.1677$$

Sample Calculation of Power Coefficient, C_P:

The following sample calculation is for the power coefficient of the fan from the 1000 rpm and 100% slider position radial fan data set. The power coefficient is a measure of the required power input of the fan. The equation shown below was used to solve for the power coefficient, C_P, where BHP is the brake horsepower, ρ is the density of the air, N is the revolutions per minute, and D is the diameter of the fan impeller.

$$C_P = \frac{BHP}{\rho N^3 D^5} \; p.3 \; Schulze \; [7] \tag{8}$$

The measured value for each input variable is shown below with units converted to metric. There were no temperatures present, so a conversion to Kelvin was unnecessary. The revolutions per minute were converted to radians per second. The values were then substituted into (8) to solve for the power coefficient.

$$BHP = 26.778\, W = 26.778\, \frac{kgm^2}{s^3}$$

$$\rho = 1.2104\, \frac{kg}{m^3}$$

$$N = 1001.1 \cancel{rpm} * \frac{2\pi\, rad}{1 \cancel{rev}} * \frac{1\, \cancel{min}}{60\, s} = 104.82 \frac{rad}{s}$$

$$D = 0.242\, m$$

$$C_P = \frac{(26.778\, \frac{\cancel{kgm^2}}{\cancel{s^3}})}{(1.2104\, \frac{\cancel{kg}}{\cancel{m^3}})(104.82\, \frac{rad}{\cancel{s}})^3 (0.242\, \cancel{m})^5}$$

$$\therefore C_P = 0.0231$$

Sample Calculation of Flow Coefficient, C_Q:

The following sample calculation is for the flow coefficient of the fan from the 1000 rpm and 100% slider position radial fan data set. The flow coefficient is a measure of the flow rate of air for the fan. The equation shown below was used to solve for the flow coefficient, C_Q, where Q is the flow rate of the air, N is the revolutions per minute, and D is the diameter of the fan impeller.

$$C_Q = \frac{Q}{ND^3}\ p.3\ Schulze\ [7] \tag{9}$$

The measured value for each input variable is shown below with units converted to metric. There were no temperatures present, so a conversion to Kelvin was unnecessary. The revolutions per minute were converted to radians per second. The values were then substituted into (9) to solve for the flow coefficient.

$$Q = 0.07928\ m^3/s$$

$$N = 1001.1\cancel{rpm} * \frac{2\pi\ rad}{1\cancel{rev}} * \frac{1\ \cancel{min}}{60\ s} = 104.82\frac{rad}{s}$$

$$D = 0.242\ m$$

$$C_Q = \frac{(0.07928\ \frac{\cancel{m^3}}{\cancel{s}})}{(104.82\frac{rad}{\cancel{s}})(0.242\ \cancel{m})^3}$$

$$\therefore C_Q = 0.0532$$

References:

[1] R. W. Fox and J. W. Mitchell, *Fox and McDonald's introduction to fluid mechanics*. Hoboken: Wiley, 2019.

[2] The Engineering ToolBox, "Individual & Universal Gas Constants: Definitions, Values, and Applications," *The Engineering Toolbox*, 2004. https://www.engineeringtoolbox.com/individual-universal-gas-constant-d_588.html

[3] K. Schulze, *Turbomachinery 1*. Gainesville, FL: 2026, pp. 4-6.

[4] TecQuipment Ltd, *MFP106 Centrifugal Fan Module User Guide*. Nottingham, UK: TecQuipment Ltd, 2012.

[5] K. Schulze, *Turbomachinery 2*. Gainesville, FL: 2026, pp. 5-7.

[6] K. Schulze, *Turbomachinery 3*. Gainesville, FL: 2026, pp. 2-7.

[7] K. Schulze, *Turbomachinery 5*. Gainesville, FL: 2026, pp. 2-3.

Table 4: Lab Report Participation Log

Name	Date	Hours Worked	Description of Tasks Performed
Bryce Addeo	2/12/2026	2	Head Rise vs Flow Rate Graph for both fans
Bryce Addeo	2/13/2026	1	BHP & WHP vs Q for 1000 rpm graph
Bryce Addeo	2/13/2026	1.5	BHP & WHP vs Q for remaining rpm graphs
Bryce Addeo	2/14/2026	1	Efficiency Curve Graph for both fans
Bryce Addeo	2/14/2026	1	Calculated Ch, Cp, and Cq for both fans
Bryce Addeo	2/15/2026	3	Ch, Cp, and Cq graphs for both fans, inserting graphs into report, & sample calc for air density
Matthew Strogach	2/15/2026	4	Experimental and calculated data charts for both fans
Matthew Strogach	2/15/2026	1	Head rise sample calculation
Sean Muller	2/16/2026	1	Flow rate sample calculation
Bryce Addeo	2/16/2026	2.5	Three paragraphs of cover letter
Matthew Strogach	2/17/2026	0.5	Calculation edits
Bryce Addeo	2/17/2026	1	Making corrections at Office Hours
Matthew Strogach	2/17/2026	0.5	Making corrections at Office Hours
Matthew Strogach	2/17/2026	2	Table edits and descriptions for the radial fan data
Matthew Strogach	2/18/2026	1.5	Descriptions for the backwards fan data tables
Bryce Addeo	2/18/2026	1	Descriptions for head rise vs flow rate graphs
Bryce Addeo	2/18/2026	1.5	Descriptions for WHP &BHP vs Q graphs
Bryce Addeo	2/18/2026	1	Descriptions for Efficiency Curves for both fans
Sean Muller	2/18/2026	1	Water horsepower sample calculation
Sean Muller	2/18/2026	1	Brake horsepower sample calculation
Sean Muller	2/19/2026	1	Efficiency sample calculation
Bryce Addeo	2/19/2026	1	Descriptions for coefficients graph for both fans
Bryce Addeo	2/19/2026	1	Results paragraph of cover letter
Sean Muller	2/19/2026	1	Head coefficient sample calculation
Sean Muller	2/20/2026	1	Power coefficient sample calculation
Sean Muller	2/20/2026	1	Flow coefficient sample calculation
Matthew Strogach	2/20/2026	1	Annual operating cost table
Sean Muller	2/20/2026	2	References
Sean Muller	2/20/2026	1	Transcribing handwritten sample calcs into report
Bryce Addeo	2/20/2026	1	Editing cover letter and coefficient sample calcs
Sean Muller	2/20/2026	1	Writing sample calculation descriptions
Sean Muller	2/21/2026	2	Correction of coefficients to rad / s from rev / s
Bryce Addeo	2/21/2026	2	Corrections, editing, checking over report
	Total Hours:	45	

Total Hours Per Group Member: Bryce: 21.5 hrs, Sean: 13 hrs, Matt: 10.5 hrs